

Evaporation

Potential and Actual Evaporation

You are given three months of daily climate data (temperature, humidity, radiation, wind) for a small catchment.

- Explain how you would estimate **potential evaporation** (E_{pt}) from this data.
- What key assumptions are you making in the calculation?
- Explain how E_{pt} relates to actual evaporation, E_t and how you might evaluate if $E_t \approx E_{pt}$ for the wet season.

Evaporation Dynamics in Soil

A lysimeter experiment was conducted in two neighbouring plots: one bare soil, one vegetated.

- Which plot would you expect to have higher total evaporation over the summer? Why?
- Discuss how **soil moisture** affects actual evaporation in both plots.
- Explain the role of **capillarity** and **transpiration** in the vegetated plot.

Vapour Pressure Deficit

Given the following measurements at midday:

Air Temperature	Relative Humidity	Saturated Vapour Pressure
25°C	40%	3.2 kPa

Table 1

- Calculate the actual vapour pressure e_a and the vapour pressure deficit (VPD).
- Briefly describe what this VPD tells us about the potential for evaporation at this time.

- What might plants do in response to this VPD?

Comparing Evaporation Measurement Methods

Evaluate the **advantages and limitations** of the following methods for measuring or estimating evaporation in a catchment:

1. Evaporation pans
2. Lysimeters
3. Water balance method (catchment scale)
4. Eddy covariance

For each method, comment on:

- The scale (point vs landscape)
- Type of surface (soil, open water, vegetation)
- Key sources of uncertainty

The Thornthwaite Method for Estimating Potential Evapotranspiration

Introduction - In hydrologic analysis, estimating evaporation is crucial but challenging. While direct measurements using lysimeters provide the most accurate data, these devices are expensive and complex to maintain. In data-sparse regions, empirical models offer practical alternatives. The **Thornthwaite method** is one of the simplest and most widely used approaches, requiring only mean monthly air temperature and geographic latitude to estimate potential evapotranspiration (E_{pt}).

The Thornthwaite equation estimates monthly potential evapotranspiration as:

$$E_{pt} = 1.62 \left(\frac{10T_a}{I} \right)^a$$

Where:

- E_{pt} = estimated potential evapotranspiration (cm/month)
- T_a = mean monthly air temperature (°C)

- I = annual heat index (dimensionless)
- a = empirically derived exponent

Step-by-Step Calculation:

1. **Compute the annual heat index** by summing across all months with $T_a > 0$:

$$I = \sum_{i=1}^{12} \left(\frac{T_{ai}}{5} \right)^{1.5}$$

2. **Calculate the exponent** using the cubic polynomial:

$$a = 0.492 + 0.0119I - 0.0000771I^2 + 0.000000675I^3$$

3. **Apply daylight correction** by multiplying monthly E_{pt} by a latitude-dependent factor to account for seasonal variations in day length.

The table below provides daylight correction factors for different latitudes:

North lat.	J	F	M	A	M	J	J	A	S	O	N	D
0E	1.04	0.94	1.04	1.01	1.04	1.01	1.04	1.04	1.01	1.04	1.01	1.04
10E	1.00	0.91	1.03	1.03	1.08	1.06	1.08	1.07	1.02	1.02	0.98	0.99
20E	0.95	0.90	1.03	1.05	1.13	1.11	1.14	1.11	1.02	1.00	0.93	0.94
30E	0.90	0.87	1.03	1.08	1.18	1.17	1.20	1.14	1.03	0.98	0.89	0.88
35E	0.87	0.85	1.03	1.09	1.21	1.21	1.23	1.16	1.03	0.97	0.86	0.85
40E	0.84	0.83	1.03	1.11	1.24	1.24	1.27	1.18	1.04	0.96	0.83	0.81
45E	0.80	0.81	1.02	1.13	1.28	1.28	1.31	1.21	1.04	0.94	0.79	0.75
50E	0.74	0.78	1.02	1.15	1.33	1.33	1.37	1.25	1.06	0.92	0.76	0.70

Table 2: NRC mean hours of bright sunshine expressed in units of 30 days of 12 hours each day

Applying the Thornthwaite Method

Problem Statement:

You are tasked with estimating the total potential evapotranspiration for a growing season in Saskatoon, Canada (latitude approximately 52°N). Using the Thornthwaite method, calculate the total E_{pt} for the period from May 16 to September 24, 1961.

Given Data:

Month	Mean Temperature (°C)
Apr	1.8
May	11.6
Jun	19.7
Jul	19.3
Aug	21.3
Sep	8.7
Oct	4.6

Table 3: Mean monthly temperatures for Saskatoon, Canada in 1961

Tasks:

1. Calculate the annual heat index I using all months with temperatures above 0°C
2. Determine the exponent a using the cubic polynomial
3. Calculate monthly E_{pt} values for May through September
4. Apply appropriate daylight correction factors for latitude 52°N
5. Sum the corrected monthly values to find the total E_{pt} for the specified period